AI & DATA SCIENCE TRACK

Sales Forecasting and Demand Prediction

Milestone 2 — Advanced Data Analysis and Feature Engineering Report

July 2026

1. Introduction

Milestone 1 produced a cleaned, store-level daily sales dataset for the Rossmann chain (cleaned_final_rossmann_data.csv: 844,392 rows, 39 columns, 1,115 stores, January 2013 to July 2015). Milestone 2 builds directly on that output. Its objective is to go deeper into the time-series structure of demand, confirm the data is statistically suitable for forecasting, engineer a richer set of predictive features, and formally rank which features matter most before Milestone 3 model development begins.

This report documents every analysis step performed, the exact statistics produced, the visualizations built, and the resulting feature engineering and selection decisions.

2. Time Series Analysis

2.1 Network-Wide Daily Sales Trend

A daily aggregate series was built by summing Sales across all open stores for each calendar day, then interpolating the handful of missing calendar dates. A 30-day rolling mean was overlaid to smooth out day-to-day noise and reveal the underlying trend.

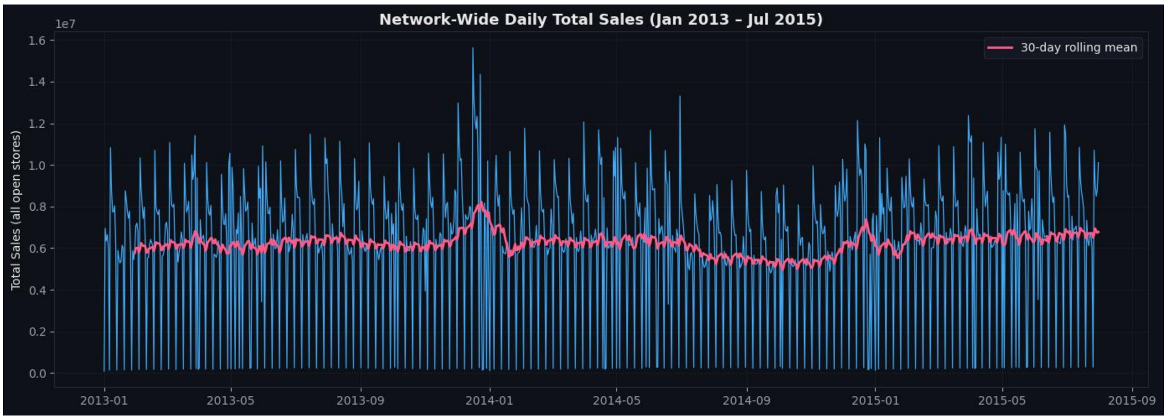

Figure 1. Network-wide daily total sales, January 2013 to July 2015, with a 30-day rolling mean overlay. The sharp daily spikes and troughs are weekly seasonality; the smoothed line reveals a rise heading into each Christmas period and a broad dip around mid-2014.

2.2 Seasonal Decomposition

An additive decomposition (period = 7, i.e. weekly) was applied to the daily series to separate it into trend, seasonal, and residual components.

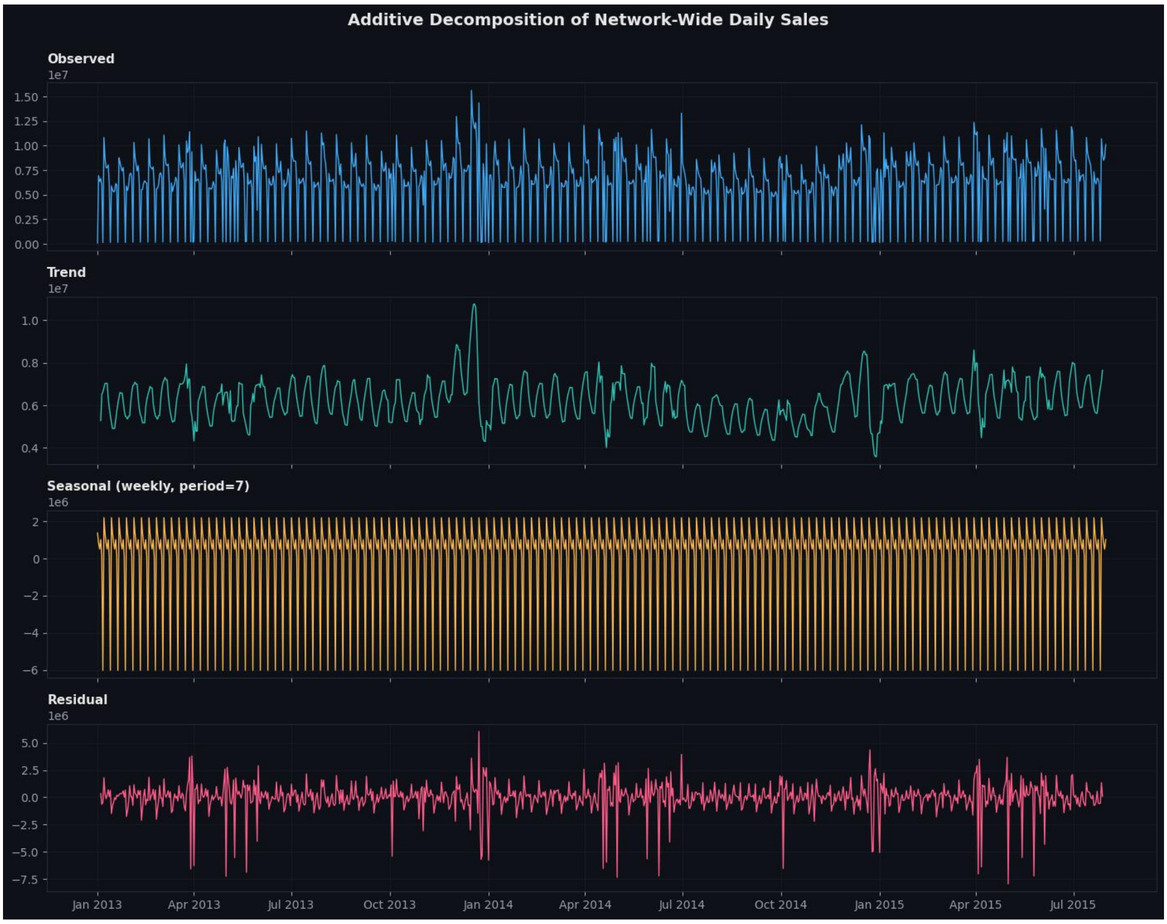

Figure 2. Additive decomposition of network-wide daily sales into Observed, Trend, Seasonal (weekly), and Residual components. The seasonal component is large and highly repeatable, confirming a strong 7-day cycle; the trend component shows a pronounced spike around the 2013-2014 holiday season.

2.3 Weekly Seasonality — Average Sales by Day of Week

Average total sales were computed per calendar weekday to quantify the weekly pattern visible in the decomposition.

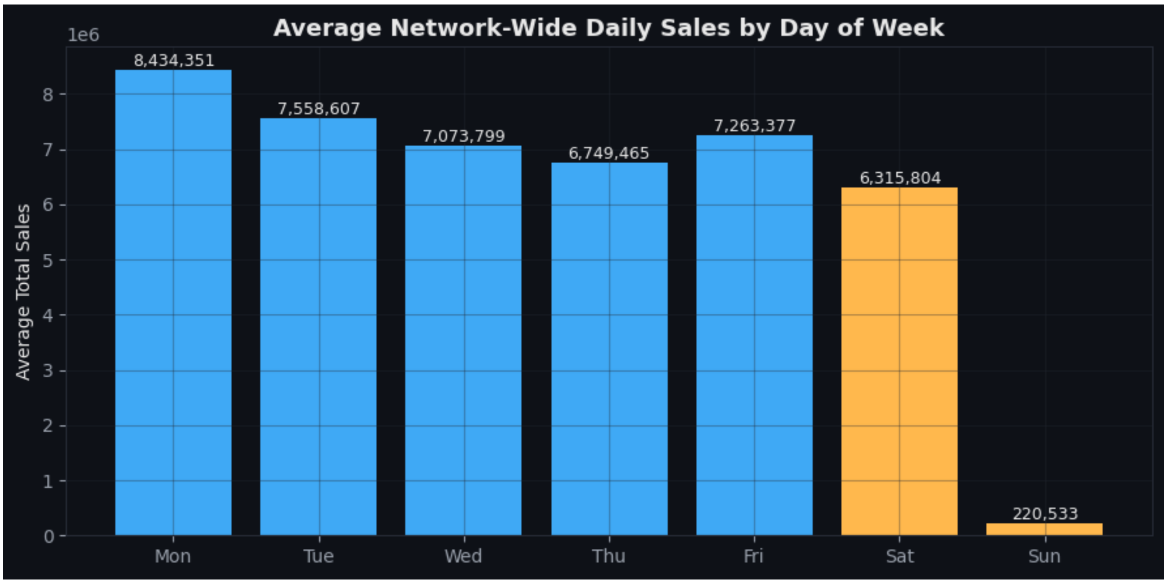

Figure 3. Average network-wide daily sales by day of week. Monday is the strongest day (8,434,351 average total sales) and Sunday the weakest by a wide margin (220,533), consistent with most stores being closed or operating minimally on Sundays.

2.4 Annual Seasonality — Monthly Totals by Year

Monthly total sales were plotted separately for each year to check whether the same seasonal shape repeats annually.

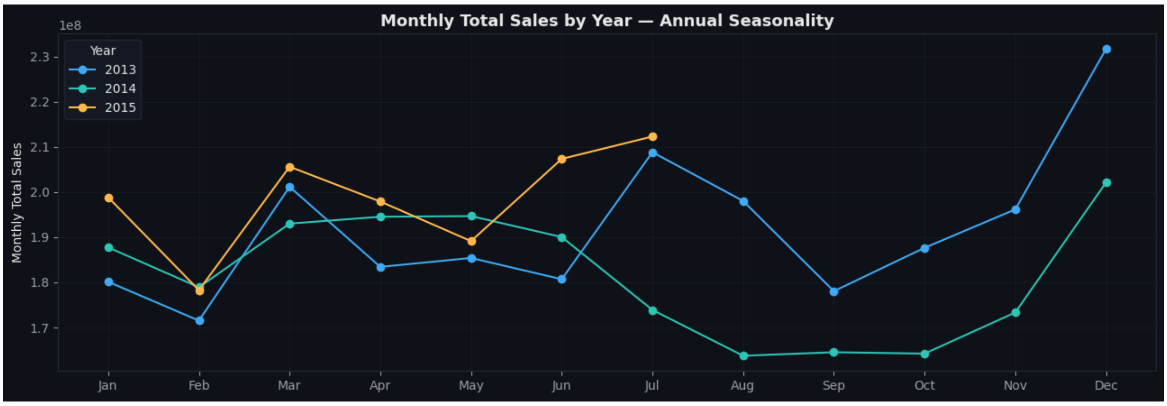

Figure 4. Monthly total sales for 2013, 2014, and 2015 overlaid by month. All three years show a February dip and a strong run-up into December, with 2013 ending in a particularly sharp December spike — evidence of a recurring, exploitable annual pattern.

3. Stationarity Testing (Augmented Dickey-Fuller)

Many time-series forecasting methods assume the series is stationary (its statistical properties do not change over time). The Augmented Dickey-Fuller (ADF) test was run on the daily Total_Sales series, both at its raw level and after first-differencing, to confirm this assumption.

Series	ADF Statistic	p-value	Lags Used	Observations	Verdict (α=0.05)
Total_Sales (level)	-4.7616	0.000064	20	921	Stationary
Total_Sales (1st diff.)	-14.0085	0.000000	19	921	Stationary

Table 1. ADF stationarity test results. Critical values at the 1%, 5%, and 10% levels were -3.4375, -2.8647, and -2.5684 respectively for both tests.

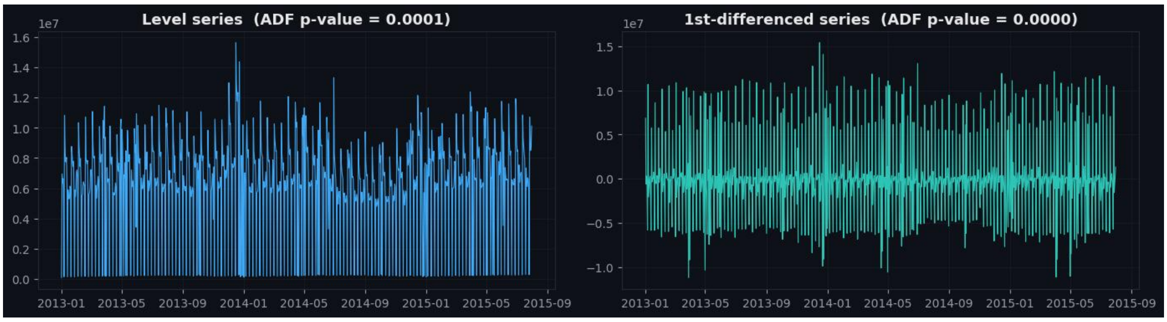

Figure 5. Level series versus 1st-differenced series, with ADF p-values annotated.

Key Finding

The level series is already stationary ($p = 0.000064$, well below 0.05), so no differencing is strictly required before modeling. The 1st-differenced series is even more strongly stationary ($p \approx 0$), confirming the series has no unit root and no long-term trend that would break standard forecasting assumptions.

4. Correlation Analysis

A Pearson correlation matrix was computed across Sales and the available engineered features from Milestone 1 (Customers, Promo, Promo2Active, holidays, CompetitionDistance, CompetitionOpen, day-of-week dummies, Month, Quarter, StoreType, Assortment, and the lag features).

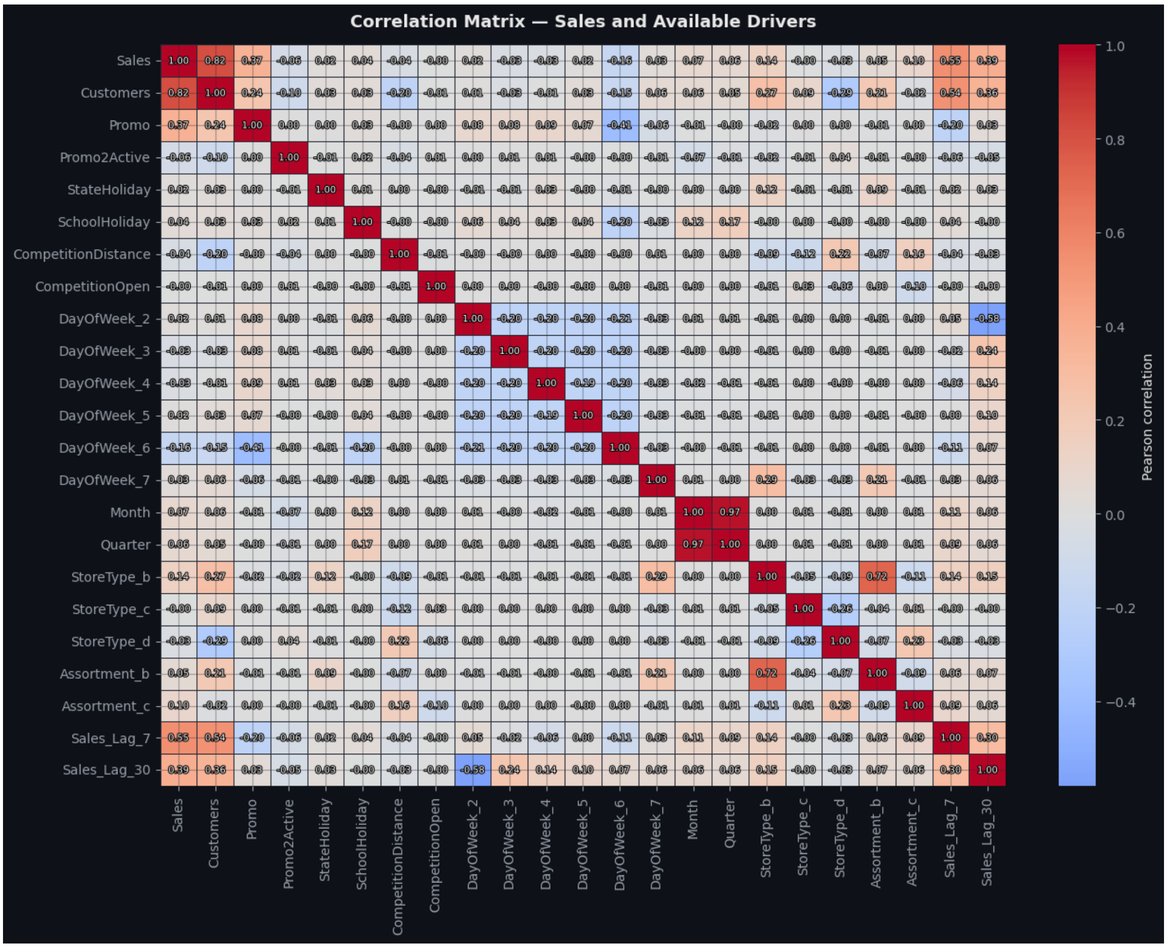

Figure 6. Correlation matrix across Sales and its available drivers.

Ranked by absolute correlation with Sales:

Feature	Correlation with Sales
Customers	0.819
Sales_Lag_7	0.546
Sales_Lag_30	0.394
Promo	0.368
DayOfWeek_6 (Saturday)	-0.158
StoreType_b	0.145
Assortment_c	0.103
Month	0.074
Promo2Active	-0.063
Quarter	0.062

Feature	Correlation with Sales
CompetitionDistance	-0.040
CompetitionOpen	-0.003

Table 2. Features ranked by absolute Pearson correlation with Sales (top 12 of 22 checked).

Key Finding

Customers is by far the strongest correlate of Sales (0.819) — unsurprising, since more shoppers in a store directly drives more transactions. Among the engineered lag features, Sales_Lag_7 (0.546) is a substantially stronger predictor than Sales_Lag_30 (0.394), meaning last week's sales carry more signal about tomorrow's sales than last month's do. CompetitionDistance and CompetitionOpen both show very weak correlation with Sales, echoing the Milestone 1 finding that competition-related fields are the least informative part of this dataset.

5. Feature Engineering

5.1 Rolling-Window Statistics

For each store, a 7-day and 30-day rolling mean of Sales were computed, along with a 7-day rolling standard deviation — all using a 1-day lag (shift(1)) so that no feature ever uses same-day information a real forecast would not have access to at prediction time.

Store	Date	Sales	RollingMean_7	RollingMean_30	RollingStd_7
1	2013-01-02	5,530	4,757.72	4,767.28	728.76
1	2013-01-05	4,997	4,781.00	4,767.28	653.51
1	2013-01-08	5,580	5,303.20	4,767.28	1,148.19
1	2013-01-10	4,892	5,366.71	5,366.71	944.27

Table 3. Sample rows showing the rolling-window features for Store 1 (values shown are pre-scaling).

5.2 Extended Lag Feature: Sales_Lag_14

Building on the Sales_Lag_7 and Sales_Lag_30 features from Milestone 1, a 14-day lag was added using the same calendar-date self-join method (rather than a row-based shift), so it correctly reflects sales from exactly two calendar weeks earlier per store. After the join and per-store fill, Sales_Lag_14 had zero remaining missing values.

5.3 Cyclical Seasonal Encodings

Month and day-of-week were each re-expressed as sine/cosine pairs (Month_sin/Month_cos, DOW_sin/DOW_cos). This lets a model recognize that December and January are adjacent, and that Sunday and Monday are adjacent — a relationship that a plain numeric encoding (1-12, or 1-7) would not capture, since it implies December and January are the farthest apart values rather than neighbors.

Date	Month	Month_sin	Month_cos	DOW_num	DOW_sin	DOW_cos	IsWeekend
2013-01-02	1	0.500	0.866	3	0.434	-0.901	0
2013-01-05	1	0.500	0.866	6	-0.782	0.623	1

Date	Month	Month_sin	Month_cos	DOW_num	DOW_sin	DOW_cos	IsWeekend
2013-01-08	1	0.500	0.866	2	0.975	-0.223	0

Table 4. Sample cyclical encoding output. IsWeekend is a simple binary derived alongside the cyclical pair for interpretability.

5.4 Aggregated Features

Two aggregation-based features were added: Store_Month_AvgSales (each store's historical average Sales for that calendar month, capturing a store-specific seasonal baseline) and Monthly_Sales_Total (the network-wide total for each calendar month, used mainly for reporting and sanity-checking rather than as a per-row model feature).

Month	Monthly_Sales_Total
2013-01	180,132,207
2013-02	171,534,275
2013-03	201,180,369
2013-04	183,431,432
2013-05	185,411,063

Table 5. Sample network-wide monthly sales totals.

5.5 Scaling the New Numeric Features

The newly engineered continuous features (Sales_RollingMean_7, Sales_RollingMean_30, Sales_RollingStd_7, Sales_Lag_14, Store_Month_AvgSales) were standardized with StandardScaler so they share the same scale as the Milestone 1 scaled features (mean 0, standard deviation 1).

Feature	Mean	Std	Min	Max
Sales_RollingMean_7	0.000	1.000	-2.454	9.323
Sales_RollingMean_30	0.000	1.000	-2.220	7.747
Sales_RollingStd_7	0.000	1.000	-1.859	15.561
Sales_Lag_14	0.000	1.000	-2.265	10.360
Store_Month_AvgSales	0.000	1.000	-1.800	7.183

Table 6. Post-scaling summary statistics for all five newly engineered continuous features (n = 844,392).

5.6 External Factors — Scope Limitation

Known Limitation

The Milestone 2 brief asks for external factors such as weather or broader economic conditions to be introduced. Consistent with the scoping note raised in the Milestone 1 report, the Rossmann dataset does not include weather or macroeconomic data of any kind — there is no field to join this from within the source files. Promotions (Promo, Promo2Active) and holidays (StateHoliday, SchoolHoliday) are the only external/contextual factors available, and both were already incorporated in Milestone 1 and reused throughout this milestone's analysis. Adding real weather data would require sourcing and joining an external dataset by store region and date, which is outside the scope of the current data source.

6. Data Visualization

6.1 Promotional Effect on Sales

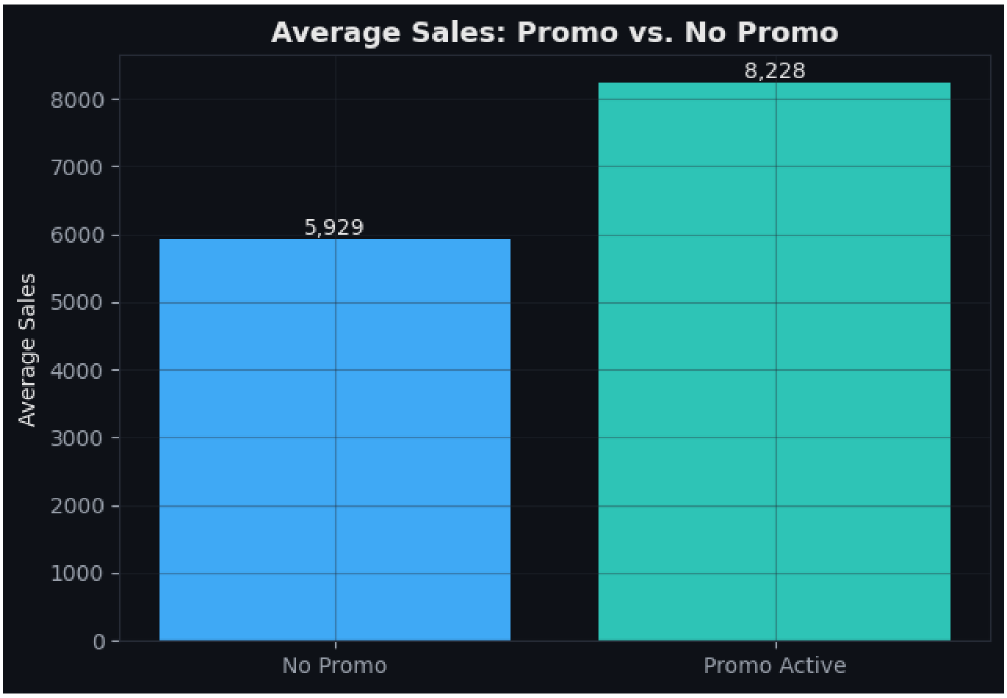

Figure 7. Average sales on non-promo days (5,929) versus promo-active days (8,228) — a 38.8% uplift.

6.2 Holiday Effects on Sales

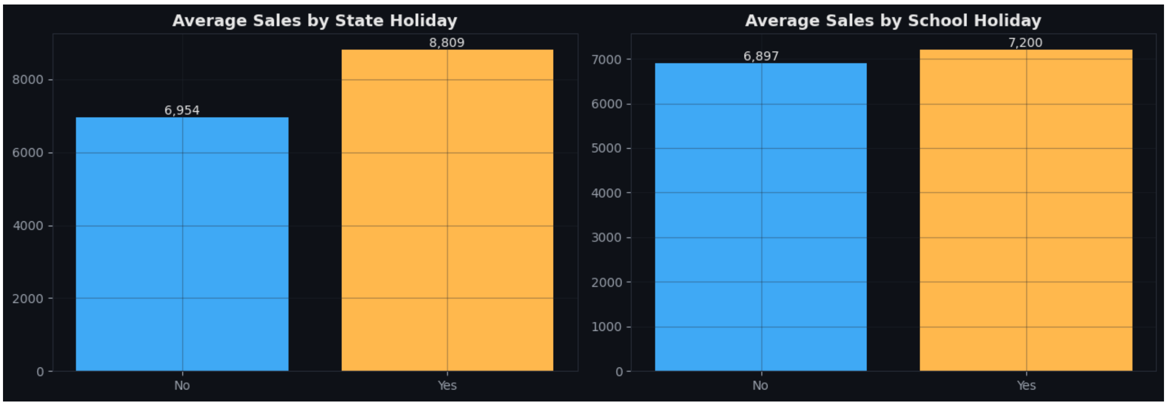

Figure 8. Average sales by state holiday status (6,954 vs. 8,809) and school holiday status (6,897 vs. 7,200). Both holiday types are associated with higher average sales rather than lower, likely reflecting the small number of stores that stay open around these days capturing concentrated demand.

6.3 Store Type and Assortment Patterns

Figure 9. Sales distribution by StoreType (b stands out with the highest median and widest spread) and by Assortment level (b/"extra" shows the highest median sales).

6.4 Average Daily Sales by Month, Split by Store Type

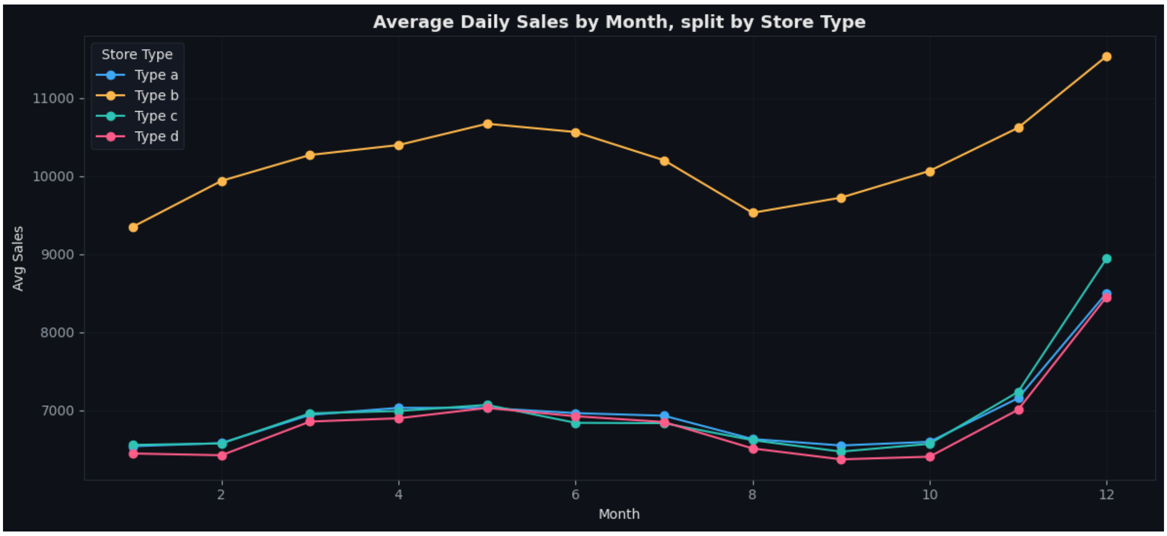

Figure 10. StoreType b runs at a consistently higher sales level across every month, with all store types showing the same December peak.

6.5 Monthly Totals and Day-of-Week Pattern

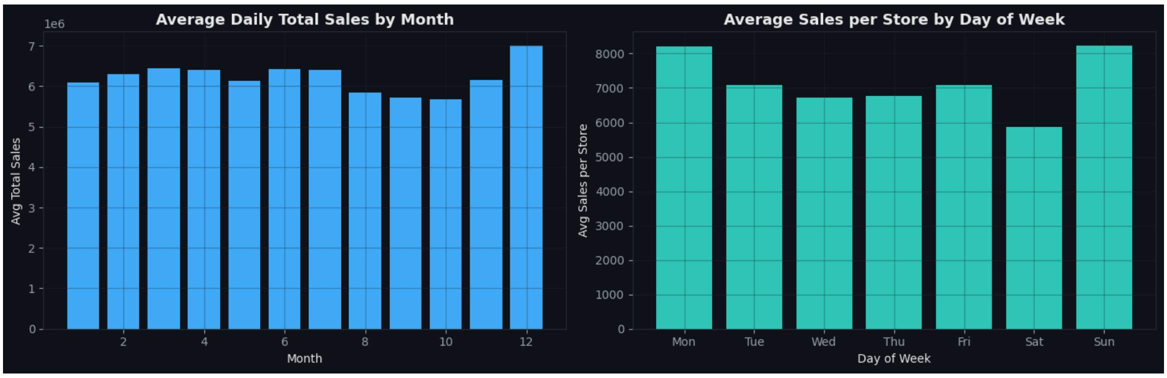

Figure 11. Left: average daily total sales by month (December highest). Right: average sales per store by day of week — Monday and Sunday are both elevated per-store on average once closed stores are excluded from the denominator, while Saturday is lowest.

6.6 Interactive Dashboards

Two interactive Plotly dashboards were built directly in the notebook:

- A network-wide historical trend chart with a 30-day rolling mean overlay, with a range slider for zooming into any period.
- A store-level explorer covering six sample stores (1, 4, 262, 500, 817, 1000), letting the viewer switch between stores via dropdown buttons to compare daily sales and each store's own 7-day rolling mean side by side.

These are interactive HTML/Plotly widgets rather than static images, so they render fully only inside the notebook itself; they are referenced here as part of the Milestone 2 deliverable rather than reproduced as static figures.

7. Feature Selection

Four independent feature-ranking methods were run against Sales on a 150,000-row sample across 31 candidate features, to cross-validate which features are genuinely predictive rather than relying on any single method's bias.

7.1 Method 1 — Pearson Correlation (Filter)

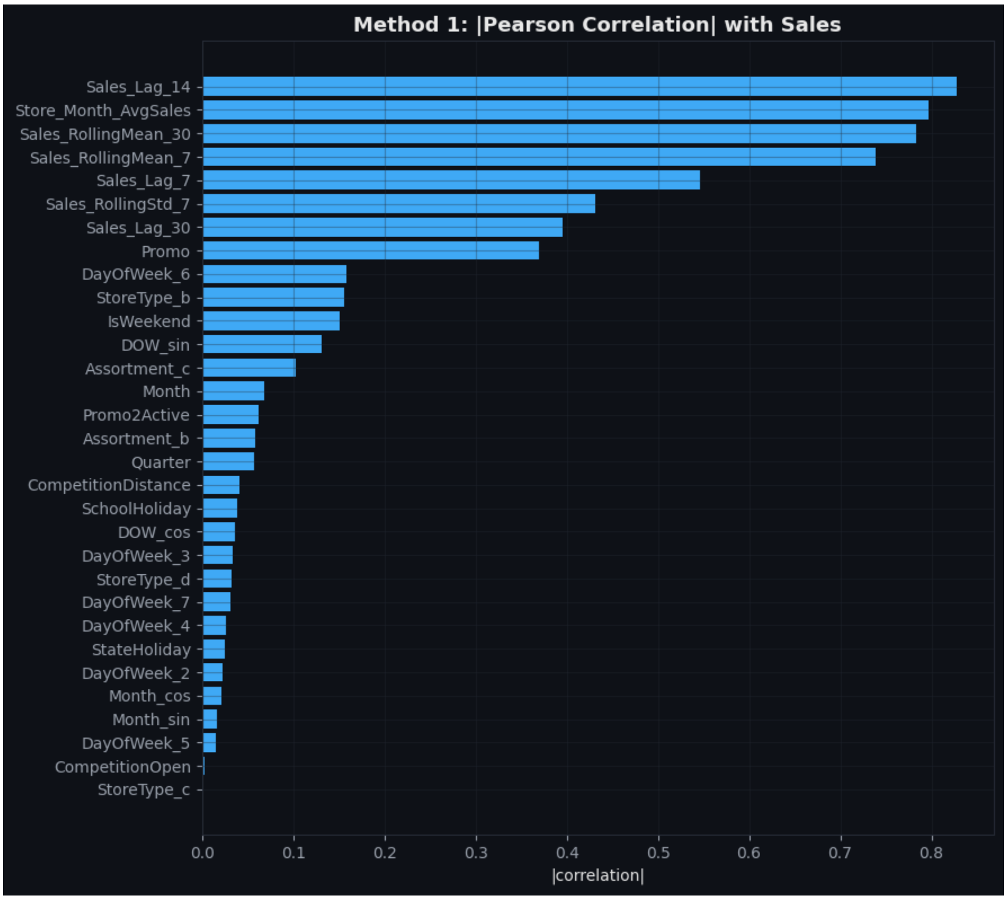

Figure 12. Absolute Pearson correlation of each candidate feature with Sales.

7.2 Method 2 — Mutual Information (Filter, information-theoretic)

Mutual information captures non-linear relationships that a linear correlation coefficient can miss. Sales_Lag_14 (0.6622), Store_Month_AvgSales (0.4859), and Sales_RollingMean_30 (0.4627) were the top three, matching the correlation-based ranking closely, plus CompetitionDistance (0.1220) ranking notably higher under this method than in the linear correlation test — suggesting distance has a non-linear relationship with Sales.

7.3 Method 3 — Random Forest Feature Importance (Embedded)

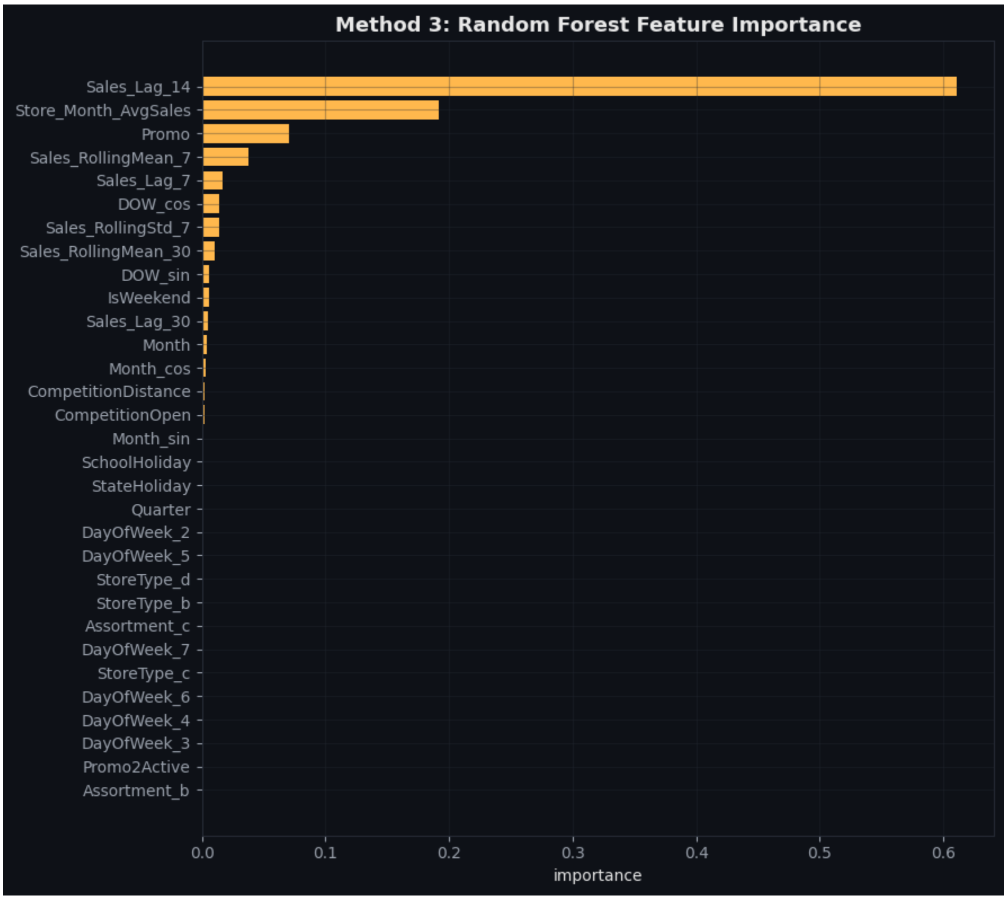

Figure 13. Random Forest feature importances (150 trees, max depth 12), trained on an 80/20 split of the 150,000-row sample.

7.4 Method 4 — Recursive Feature Elimination (Wrapper)

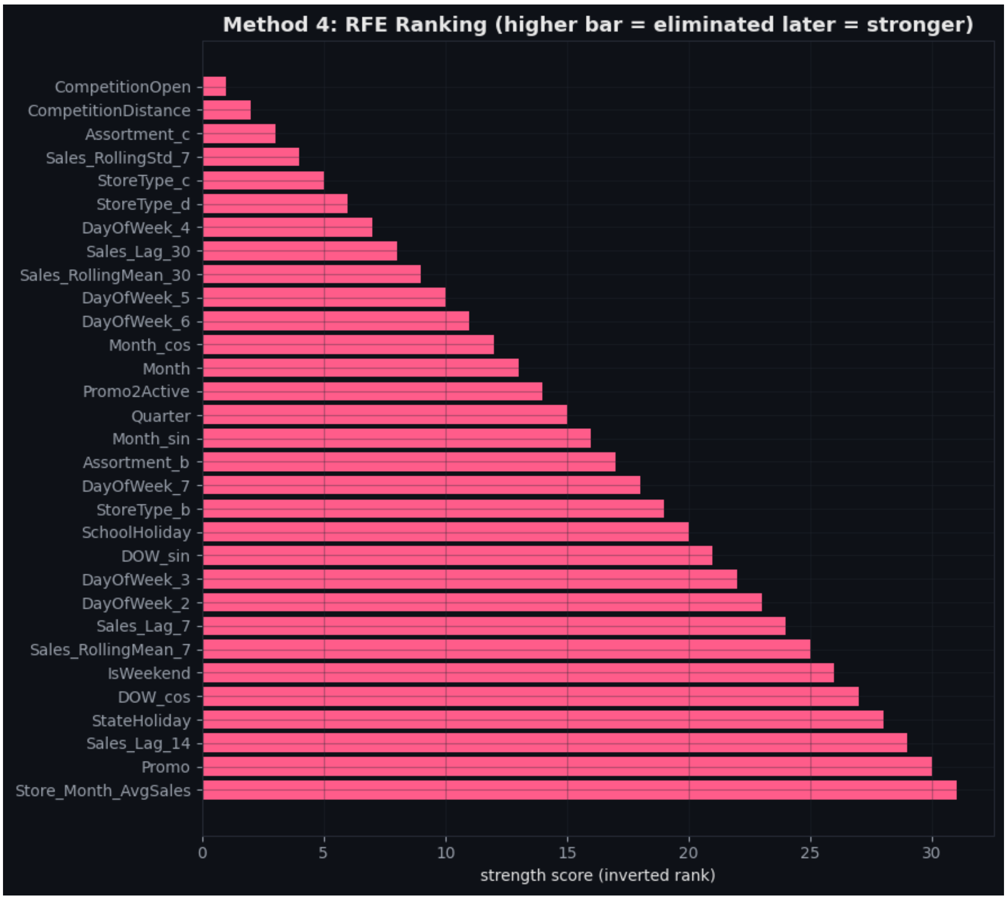

Figure 14. RFE ranking using a Ridge regression estimator (higher bar = eliminated later = stronger feature). Store_Month_AvgSales and Promo were the last two features standing.

7.5 Consensus Feature Set

The four rankings were averaged into a single consensus score per feature (lower average rank = stronger overall).

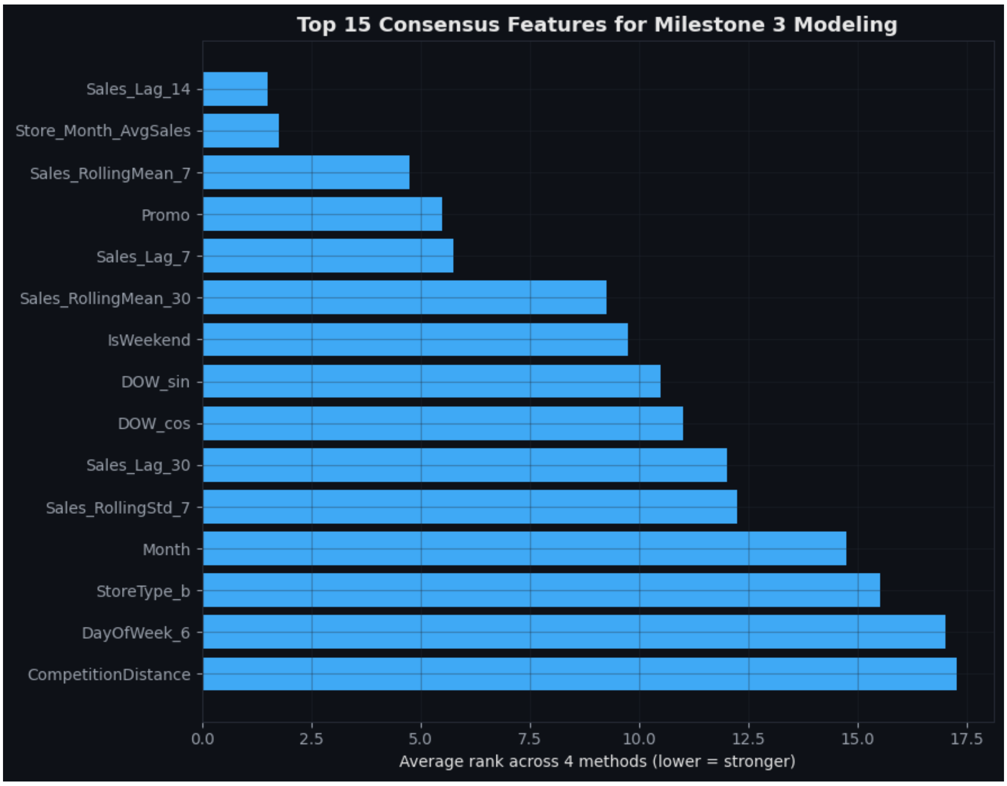

Figure 15. Top 15 consensus features for Milestone 3 modeling, ranked by average rank across all four methods.

Rank	Feature	Avg. Rank
1	Sales_Lag_14	1.50
2	Store_Month_AvgSales	1.75
3	Sales_RollingMean_7	4.75
4	Promo	5.50
5	Sales_Lag_7	5.75
6	Sales_RollingMean_30	9.25
7	IsWeekend	9.75
8	DOW_sin	10.50
9	DOW_cos	11.00
10	Sales_Lag_30	12.00
11	Sales_RollingStd_7	12.25
12	Month	14.75
13	StoreType_b	15.50
14	DayOfWeek_6	17.00
15	CompetitionDistance	17.25

Table 7. Top 15 consensus-ranked features (of 31 candidates evaluated), to be carried forward into Milestone 3 modeling.

Key Finding

Sales_Lag_14 and Store_Month_AvgSales rank 1st and 2nd across every single method — an unusually strong and consistent signal. This suggests that a two-week lag captures the right amount of recent momentum without being as noisy as a 7-day lag or as stale as a 30-day lag, and that a store's own historical seasonal baseline for the month is highly predictive on its own. CompetitionDistance and CompetitionOpen remain weak across all four methods, reinforcing that competition-related fields are the least valuable part of this dataset for forecasting.

8. Key Insights and Recommendations for Modeling

- The daily Sales series is already stationary at the level, so Milestone 3 models are not required to difference the series first, though differencing remains a safe option since it is even more strongly stationary.
- Customers and the two shorter lag features (Sales_Lag_7, Sales_Lag_14) are consistently the strongest individual predictors — any Milestone 3 model should treat these as core inputs.
- Promo shows a clear, consistent 38.8% average sales uplift and ranks in the top 5 across every feature-selection method — it should always be included regardless of which final feature subset is chosen.
- The top 15 consensus features in Table 7 are recommended as the starting feature set for Milestone 3, with the remaining 16 candidates available as a fallback if model performance needs improvement.
- Weather and macroeconomic data remain unavailable in this dataset (Section 5.6); this should be re-confirmed as an accepted scope limitation before Milestone 3, since it directly affects how far demand-forecast accuracy can realistically be pushed.

9. Challenges Encountered

Several non-trivial challenges came up during this milestone, beyond routine analysis and feature engineering:

Challenge 1 — Preventing Look-Ahead Leakage in Rolling and Lag Features

Every rolling-window feature (Sales_RollingMean_7/30, Sales_RollingStd_7) had to be built with an explicit shift(1) before the rolling window, so that the value available for a given day never includes that day's own sales. It is easy to accidentally compute a rolling mean that includes the current row, which would silently leak future information into the training data and make offline model performance look far better than it would in real deployment.

Challenge 2 — Reconciling Disagreement Between Feature-Selection Methods

The four feature-selection methods did not always agree. CompetitionDistance, for example, ranked poorly under linear Pearson correlation (rank 18) but noticeably better under Mutual Information (rank 7) and Random Forest (rank 14), since it captures a non-linear relationship that a linear correlation coefficient cannot see. Deciding on a single consensus feature set required averaging ranks across all four methods rather than trusting any one method alone, since each has known blind spots (linear-only, sample-size sensitivity, or greedy elimination order).

Challenge 3 — Computational Cost of Wrapper and Embedded Methods

Running Random Forest importance and RFE (a wrapper method that repeatedly refits a model while removing one feature at a time) on the full 844,392-row dataset would have been very slow, especially for RFE which refits once per feature. A 150,000-row random sample was used for the feature-selection stage specifically, while the earlier time-series and correlation analyses used the full dataset — this distinction is documented in this report so the two are not confused.

Challenge 4 — Keeping Two Parallel Day-of-Week Encodings Consistent

Milestone 1 already one-hot encoded DayOfWeek into DayOfWeek_2 through DayOfWeek_7. Milestone 2 additionally needed a cyclical sine/cosine encoding of the same information for the correlation and feature-selection stages. Reconstructing a single numeric DayOfWeek_num from the six dummy columns (rather than re-deriving it from Date) required care to make sure Monday, which has no dummy column of its own under drop_first=True one-hot encoding, was correctly inferred as the default case.

Challenge 5 — Weather and Economic Data Remain Unavailable

As already flagged in Milestone 1 and re-confirmed in Section 5.6, the Rossmann dataset has no weather or macroeconomic fields to draw on. This is a data-availability constraint rather than an analysis gap, but it does mean the "external factors" task in the brief could only be addressed through the promotional and holiday fields already present, not through genuinely new external data sources.

10. Conclusion

Milestone 2 confirmed the dataset's suitability for time-series forecasting (stationary at the level, strong and repeatable weekly and annual seasonality), engineered five new predictive features on top of Milestone 1's output, and used four independent feature-selection methods to converge on a consensus ranking. Sales_Lag_14 and Store_Month_AvgSales emerged as the two strongest features by a clear margin. With this ranked feature set and a fully documented analysis trail, the project is ready to proceed into Milestone 3 model development.